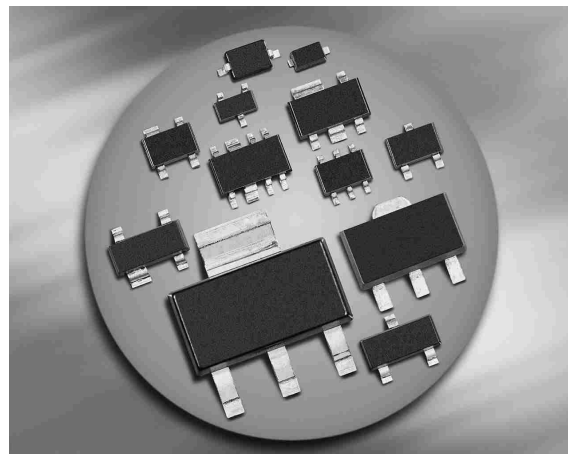
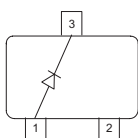


## Silicon Schottky Diodes

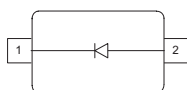
- For low-loss, fast-recovery, meter protection, bias isolation and clamping application
- Integrated diffused guard ring
- Low forward voltage
- Pb-free (RoHS compliant) package
- Qualified according AEC Q101



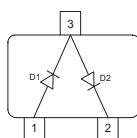
### BAT64



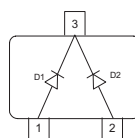
### BAT64-02W BAT64-02V



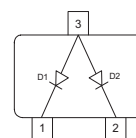
### BAT64-04 BAT64-04W



### BAT64-05 BAT64-05W



### BAT64-06 BAT64-06W



**ESD (Electrostatic discharge) sensitive device, observe handling precaution!**

| Type       | Package | Configuration  | $L_S$ (nH) | Marking |
|------------|---------|----------------|------------|---------|
| BAT64      | SOT23   | single         | 1.8        | 63s     |
| BAT64-02V  | SC79    | single         | 0.6        | t       |
| BAT64-02W* | SCD80   | single         | 0.6        | 64      |
| BAT64-04   | SOT23   | series         | 1.8        | 64s     |
| BAT64-04W  | SOT323  | series         | 1.4        | 64s     |
| BAT64-05   | SOT23   | common cathode | 1.8        | 65s     |
| BAT64-05W  | SOT323  | common cathode | 1.4        | 65s     |
| BAT64-06   | SOT23   | common anode   | 1.8        | 66s     |
| BAT64-06W  | SOT323  | common anode   | 1.4        | 66s     |

\* Not for new design

**Maximum Ratings** at  $T_A = 25\text{ °C}$ , unless otherwise specified

| Parameter                                                             | Symbol    | Value       | Unit |
|-----------------------------------------------------------------------|-----------|-------------|------|
| Diode reverse voltage                                                 | $V_R$     | 40          | V    |
| Forward current                                                       | $I_F$     | 250         | mA   |
| Non-repetitive peak surge forward current<br>( $t \leq 10\text{ms}$ ) | $I_{FSM}$ | 800         |      |
| Average rectified forward current (50/60Hz, sinus)                    | $I_{FAV}$ | 120         |      |
| Total power dissipation                                               | $P_{tot}$ |             | mW   |
| BAT64, $T_S \leq 86\text{°C}$                                         |           | 250         |      |
| BAT64-02W, -02V $T_S \leq 121\text{°C}$                               |           | 250         |      |
| BAT64-04, BAT64-06, $T_S \leq 61\text{°C}$                            |           | 250         |      |
| BAT64-04W, BAT64-06W, $T_S \leq 111\text{°C}$                         |           | 250         |      |
| BAT64-05, $T_S \leq 36\text{°C}$                                      |           | 250         |      |
| BAT64-05W, $T_S \leq 104\text{°C}$                                    |           | 250         |      |
| Junction temperature                                                  | $T_j$     | 150         | °C   |
| Storage temperature                                                   | $T_{stg}$ | -55 ... 150 |      |

### Thermal Resistance

| Parameter                                | Symbol     | Value      | Unit |
|------------------------------------------|------------|------------|------|
| Junction - soldering point <sup>1)</sup> | $R_{thJS}$ |            | K/W  |
| BAT64                                    |            | $\leq 255$ |      |
| BAT64-02W, -02V                          |            | $\leq 115$ |      |
| BAT64-04, BAT64-06,                      |            | $\leq 355$ |      |
| BAT64-04W, BAT64-06W                     |            | $\leq 155$ |      |
| BAT64-05                                 |            | $\leq 455$ |      |
| BAT64-05W                                |            | $\leq 185$ |      |

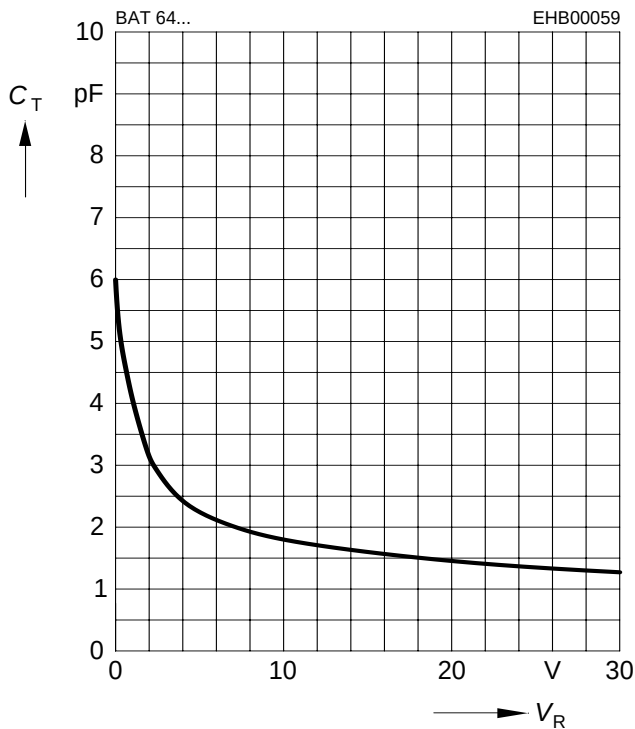
<sup>1)</sup>For calculation of  $R_{thJA}$  please refer to Application Note AN077 (Thermal Resistance Calculation)

**Electrical Characteristics** at  $T_A = 25\text{ °C}$ , unless otherwise specified

| Parameter                                                                                                                       | Symbol     | Values                   |                          |                          | Unit          |
|---------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|--------------------------|--------------------------|---------------|
|                                                                                                                                 |            | min.                     | typ.                     | max.                     |               |
| DC Characteristics                                                                                                              |            |                          |                          |                          |               |
| Breakdown voltage<br>$I_{(BR)} = 10\text{ }\mu\text{A}$                                                                         | $V_{(BR)}$ | 40                       | -                        | -                        | V             |
| Reverse current<br>$V_R = 30\text{ V}$<br>$V_R = 30\text{ V}, T_A = 85\text{ }^{\circ}\text{C}$                                 | $I_R$      | -<br>-                   | -<br>-                   | 2<br>200                 | $\mu\text{A}$ |
| Forward voltage<br>$I_F = 1\text{ mA}$<br>$I_F = 10\text{ mA}$<br>$I_F = 30\text{ mA}$<br>$I_F = 100\text{ mA}$                 | $V_F$      | 270<br>310<br>370<br>500 | 320<br>385<br>440<br>570 | 350<br>430<br>520<br>750 | mV            |
| AC Characteristics                                                                                                              |            |                          |                          |                          |               |
| Diode capacitance<br>$V_R = 1\text{ V}, f = 1\text{ MHz}$                                                                       | $C_T$      | -                        | 4                        | 6                        | pF            |
| Reverse recovery time<br>$I_F = 10\text{ mA}, I_R = 10\text{ mA}$ , measured $I_R = 1\text{ mA}$ ,<br>$R_L = 100\text{ }\Omega$ | $t_{rr}$   | -                        | -                        | 5                        | ns            |

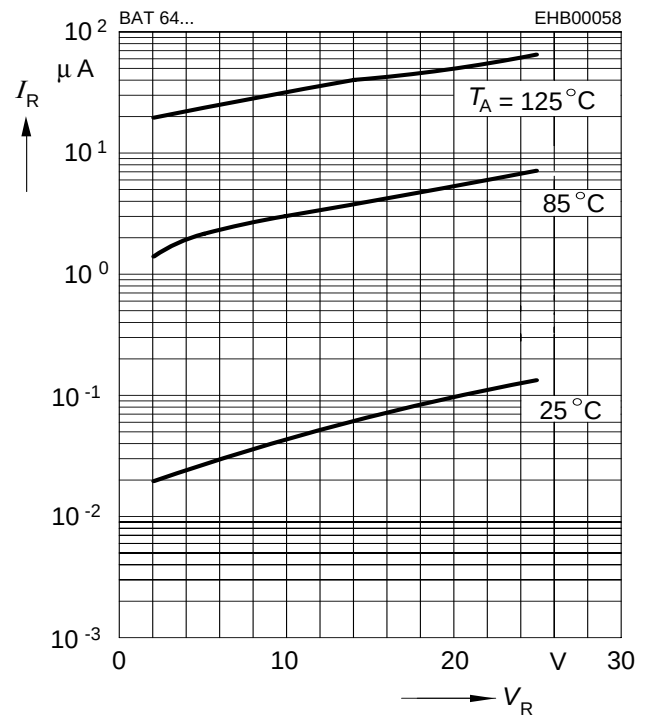
**Diode capacitance**  $C_T = f(V_R)$

$f = 1\text{MHz}$



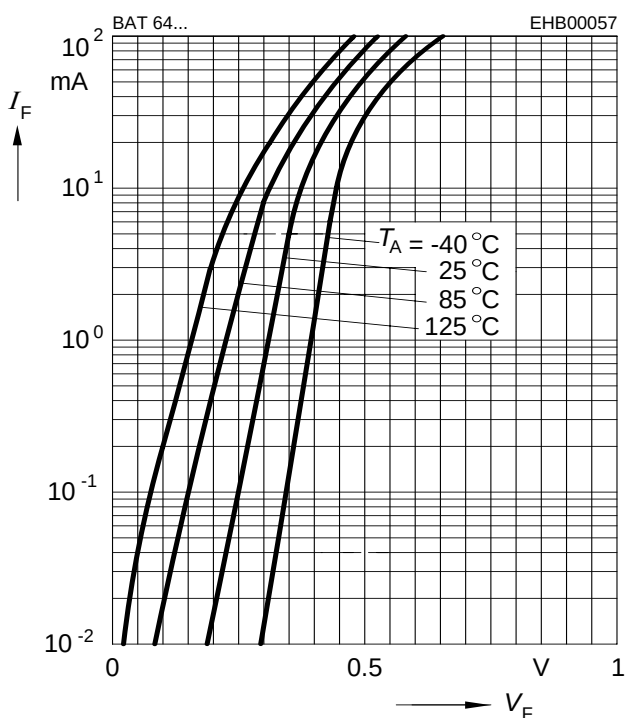
**Reverse current**  $I_R = f(V_R)$

$T_A = \text{Parameter}$



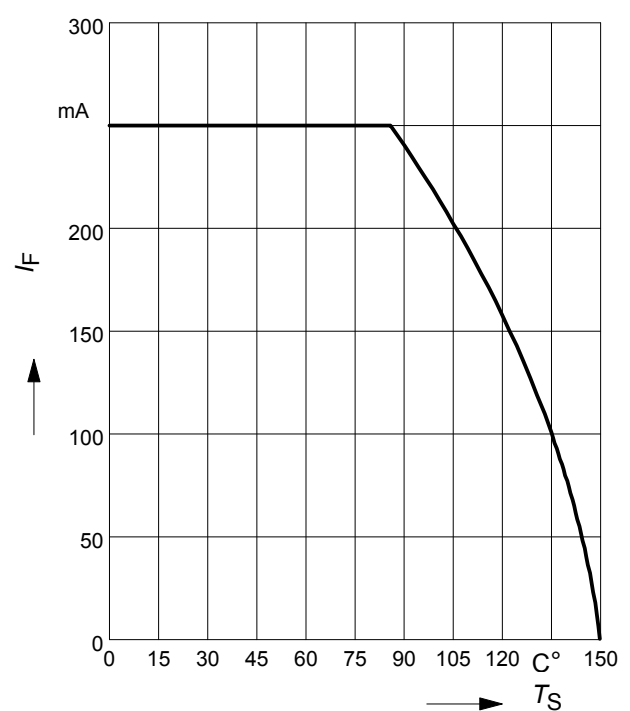
**Forward current**  $I_F = f(V_F)$

$T_A = \text{Parameter}$



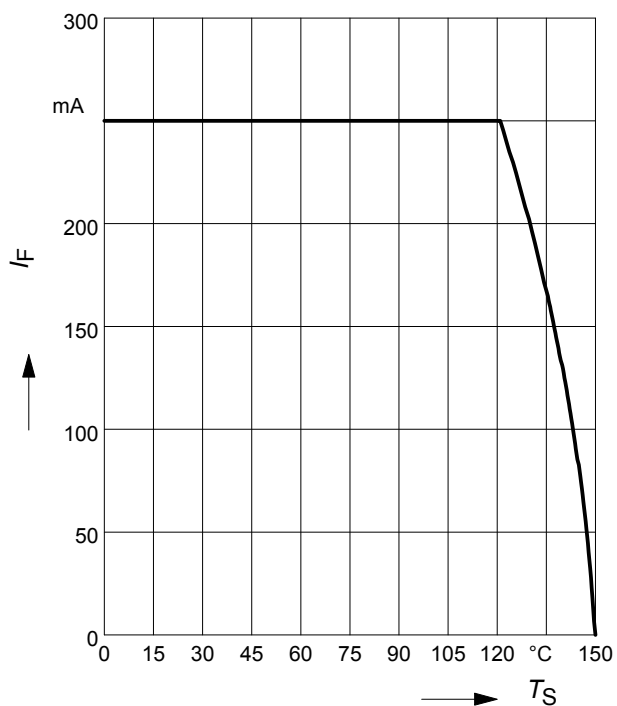
**Forward current**  $I_F = f(T_S)$

BAT64



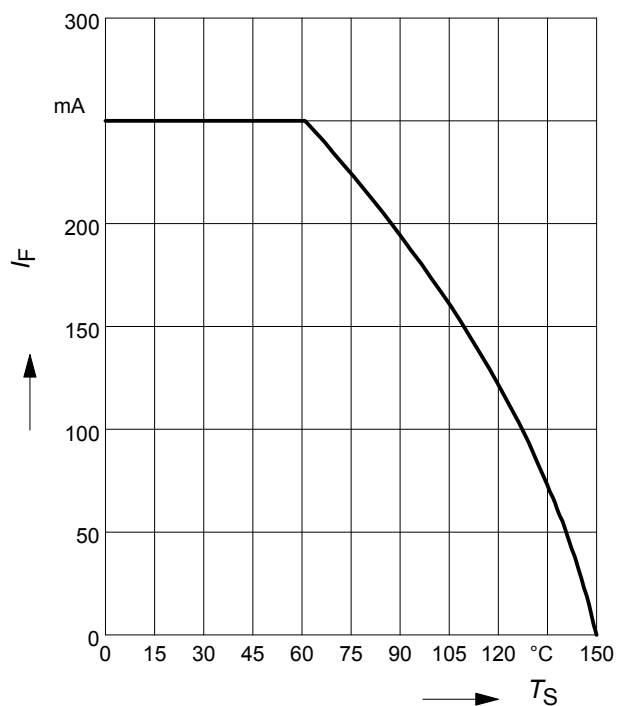
**Forward current  $I_F = f(T_S)$**

BAT64-02W, -02V



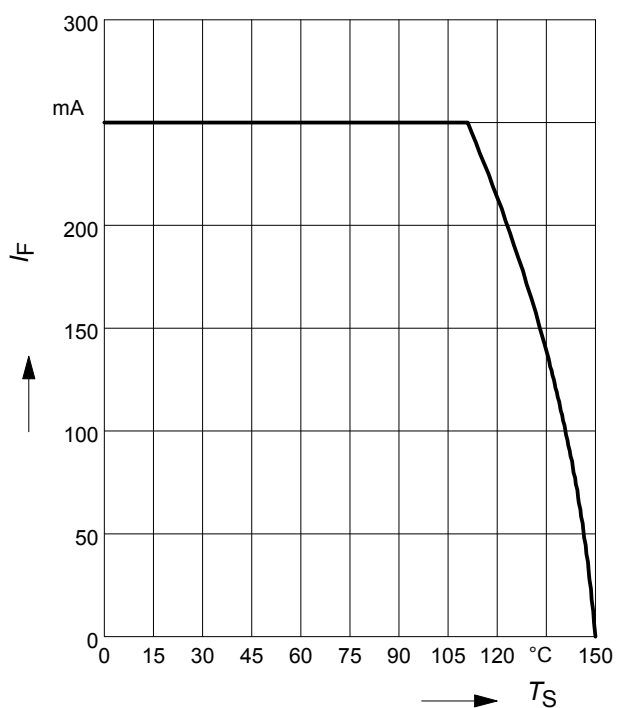
**Forward current  $I_F = f(T_S)$**

BAT64-04, BAT64-06



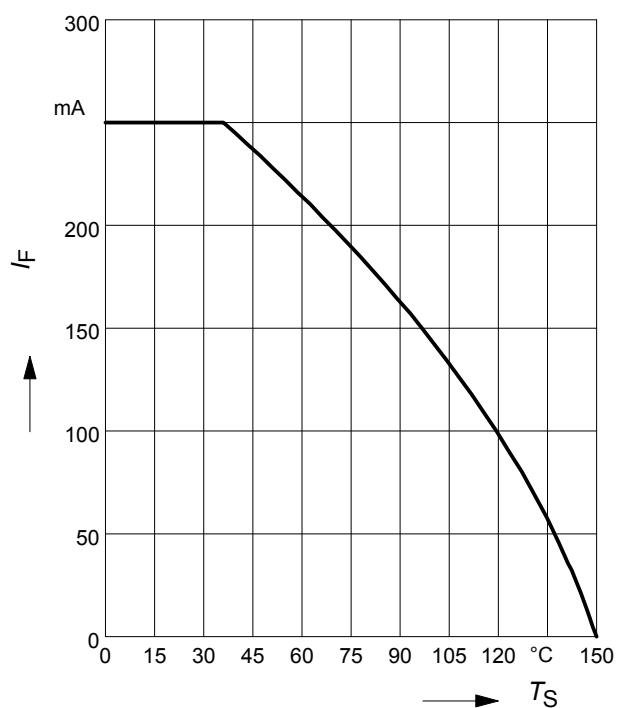
**Forward current  $I_F = f(T_S)$**

BAT64-04W, BAT64-06W



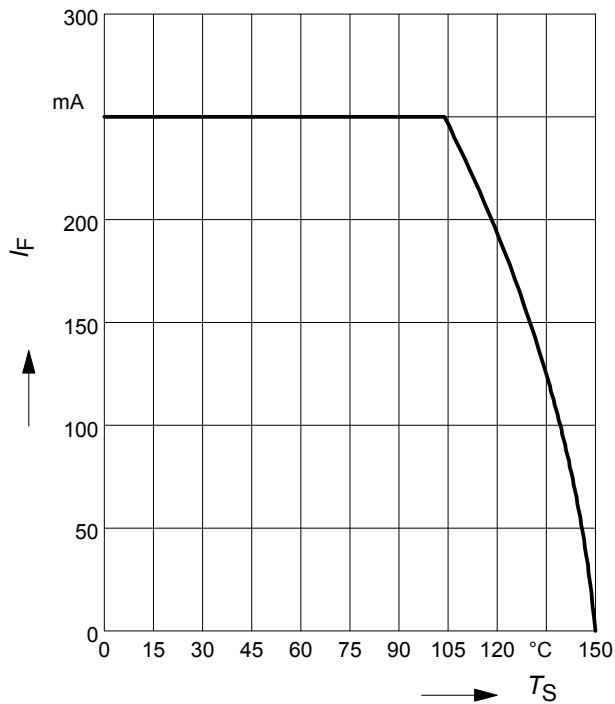
**Forward current  $I_F = f(T_S)$**

BAT64-05



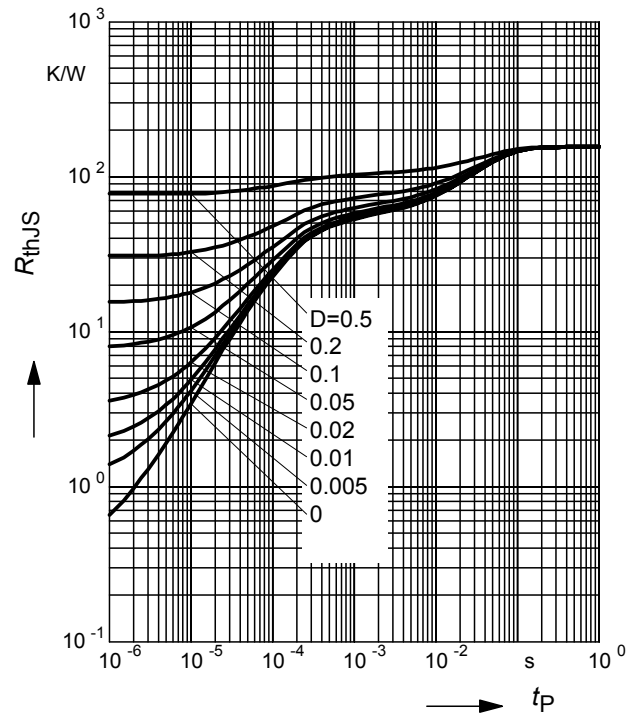
### Forward current $I_F = f(T_S)$

BAT64-05W



### Permissible Puls Load $R_{thJS} = f(t_p)$

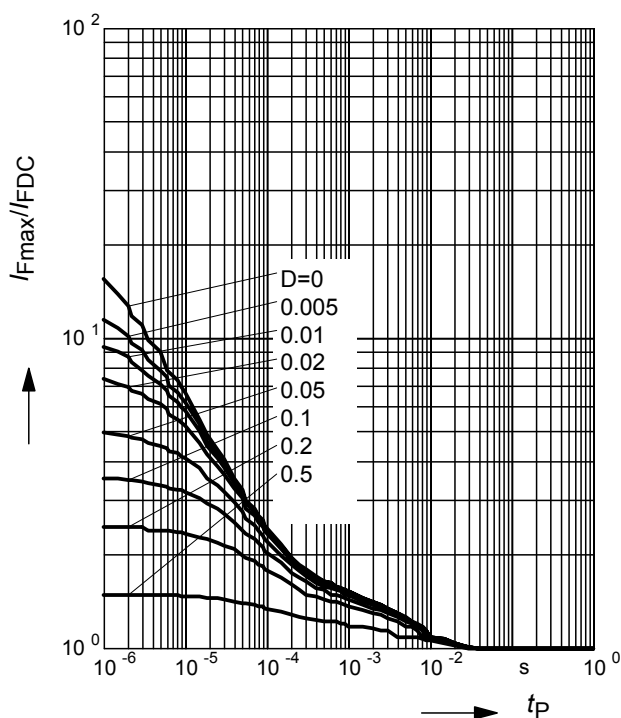
BAT64-02W, -02V



### Permissible Pulse Load

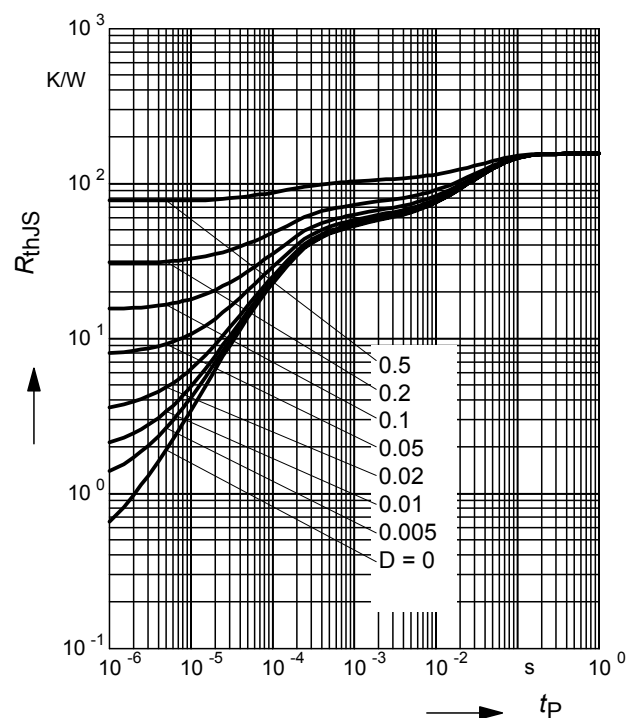
$I_{Fmax}/I_{FDC} = f(t_p)$

BAT64-02W, -02V



### Permissible Puls Load $R_{thJS} = f(t_p)$

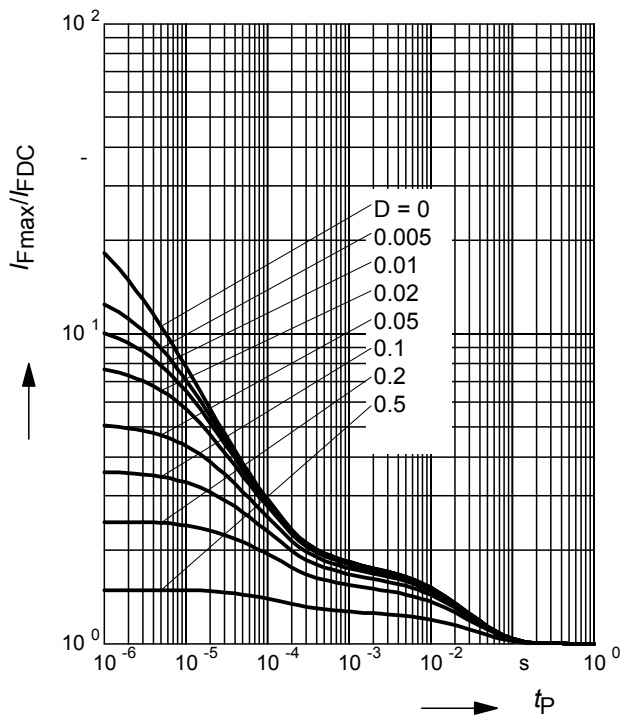
BAT64-04W, BAT64-06W



### Permissible Pulse Load

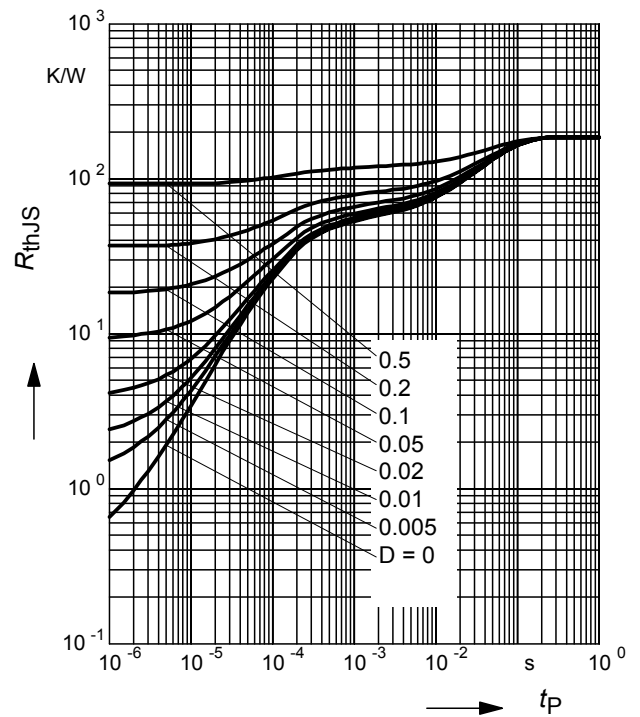
$$I_{Fmax}/I_{FDC} = f(t_p)$$

BAT64-04W, BAT64-06W



### Permissible Puls Load $R_{thJS} = f(t_p)$

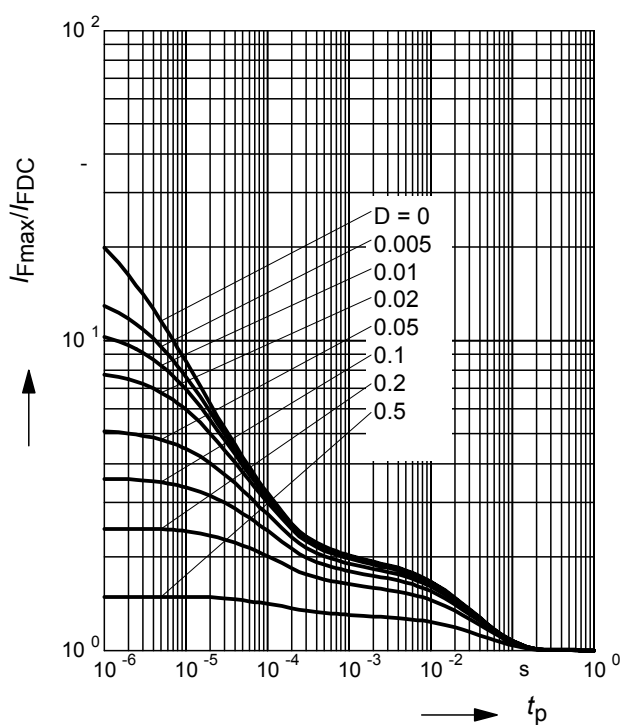
BAT64-05W



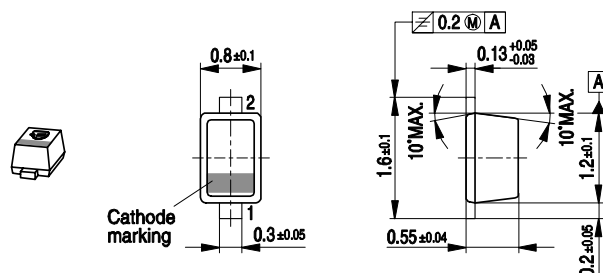
### Permissible Pulse Load

$$I_{Fmax}/I_{FDC} = f(t_p)$$

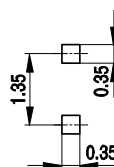
BAT64-05W



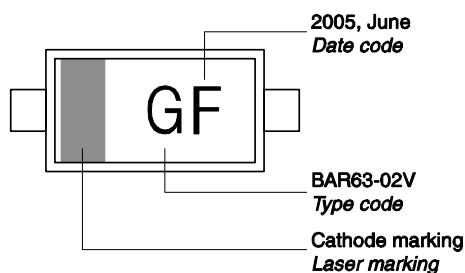
## Package Outline



## Foot Print

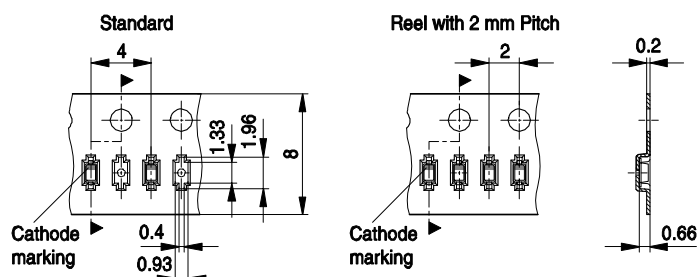


## Marking Layout (Example)

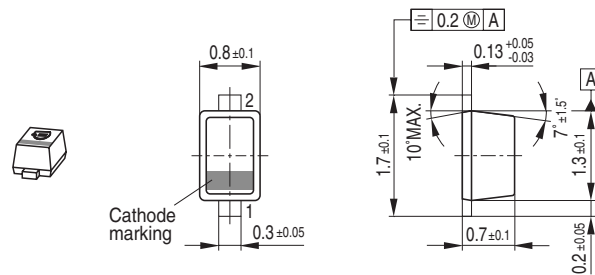


## Standard Packing

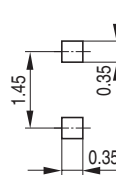
Reel ø180 mm = 3.000 Pieces/Reel  
 Reel ø180 mm = 8.000 Pieces/Reel (2 mm Pitch)  
 Reel ø330 mm = 10.000 Pieces/Reel



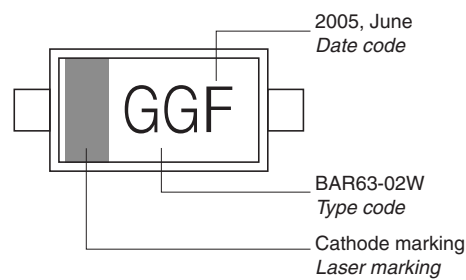
## Package Outline



## Foot Print

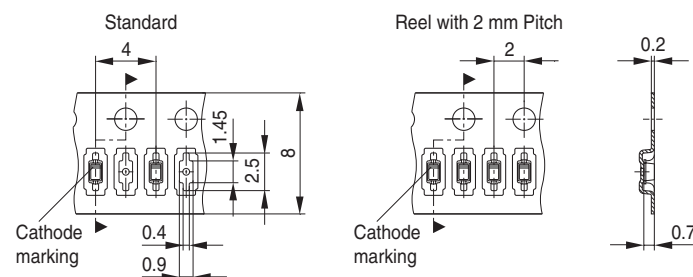


## Marking Layout (Example)



## Standard Packing

Reel ø180 mm = 3.000 Pieces/Reel  
 Reel ø180 mm = 8.000 Pieces/Reel (2 mm Pitch)  
 Reel ø330 mm = 10.000 Pieces/Reel

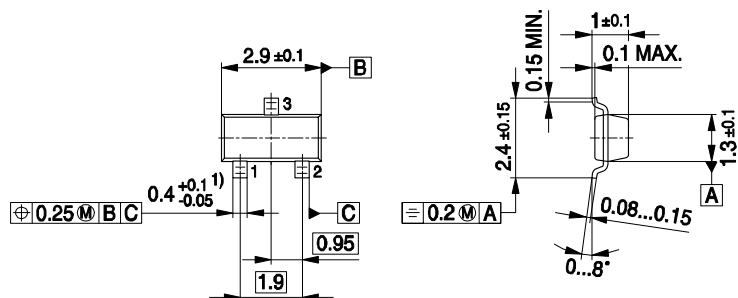


Date Code marking for discrete packages with one digit (SCD80, SC79, SC75<sup>1)</sup>) CES-Code

| Month | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 |
|-------|------|------|------|------|------|------|------|------|------|------|------|------|
| 01    | a    | p    | A    | P    | a    | p    | A    | P    | a    | p    | A    | P    |
| 02    | b    | q    | B    | Q    | b    | q    | B    | Q    | b    | q    | B    | Q    |
| 03    | c    | r    | C    | R    | c    | r    | C    | R    | c    | r    | C    | R    |
| 04    | d    | s    | D    | S    | d    | s    | D    | S    | d    | s    | D    | S    |
| 05    | e    | t    | E    | T    | e    | t    | E    | T    | e    | t    | E    | T    |
| 06    | f    | u    | F    | U    | f    | u    | F    | U    | f    | u    | F    | U    |
| 07    | g    | v    | G    | V    | g    | v    | G    | V    | g    | v    | G    | V    |
| 08    | h    | x    | H    | X    | h    | x    | H    | X    | h    | x    | H    | X    |
| 09    | j    | y    | J    | Y    | j    | y    | J    | Y    | j    | y    | J    | Y    |
| 10    | k    | z    | K    | Z    | k    | z    | K    | Z    | k    | z    | K    | Z    |
| 11    | l    | 2    | L    | 4    | l    | 2    | L    | 4    | l    | 2    | L    | 4    |
| 12    | n    | 3    | N    | 5    | n    | 3    | N    | 5    | n    | 3    | N    | 5    |

1) New Marking Layout for SC75, implemented at October 2005.

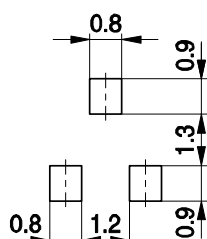
## Package Outline



1) Lead width can be 0.6 max. in dambar area

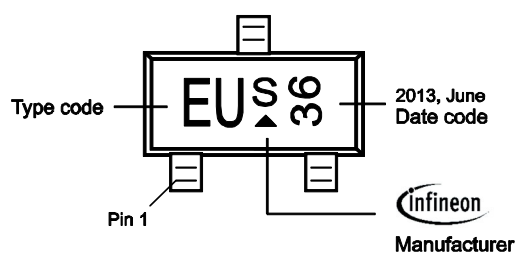
SOT23-PO V08

## Foot Print



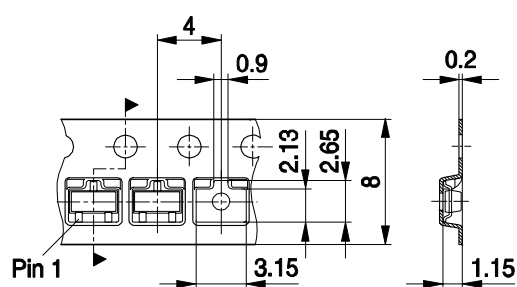
SOT23-FPR V08

## Marking Layout



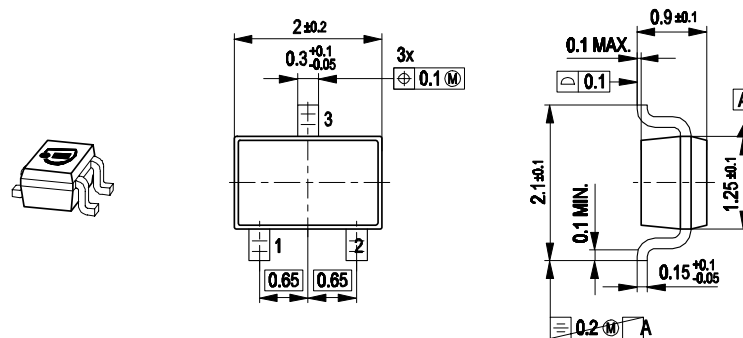
## Standard Packing

Reel o 180 mm: 3.000 Pieces / Reel  
Reel o 330 mm = 10.000 Pieces / Reel

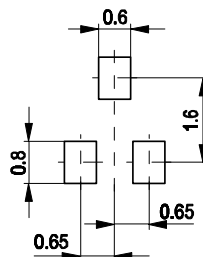


SOT23-TP V02

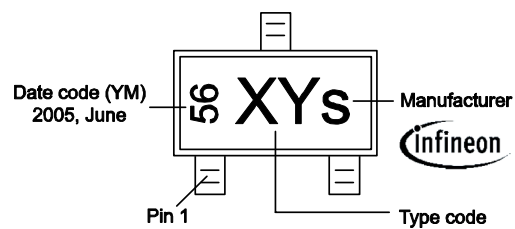
## Package Outline



## Foot Print

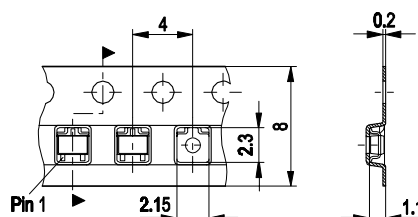


## Marking Layout (Example)



## Standard Packing

Reel  $\varnothing 180$  mm = 3.000 Pieces/Reel  
 Reel  $\varnothing 330$  mm = 10.000 Pieces/Reel



**Edition 2009-11-16**

**Published by  
Infineon Technologies AG  
81726 Munich, Germany**

**© 2009 Infineon Technologies AG  
All Rights Reserved.**

## **Legal Disclaimer**

The information given in this document shall in no event be regarded as a guarantee of conditions or characteristics. With respect to any examples or hints given herein, any typical values stated herein and/or any information regarding the application of the device, Infineon Technologies hereby disclaims any and all warranties and liabilities of any kind, including without limitation, warranties of non-infringement of intellectual property rights of any third party.

## **Information**

For further information on technology, delivery terms and conditions and prices, please contact the nearest Infineon Technologies Office ([<www.infineon.com>](http://www.infineon.com)).

## **Warnings**

Due to technical requirements, components may contain dangerous substances. For information on the types in question, please contact the nearest Infineon Technologies Office.

Infineon Technologies components may be used in life-support devices or systems only with the express written approval of Infineon Technologies, if a failure of such components can reasonably be expected to cause the failure of that life-support device or system or to affect the safety or effectiveness of that device or system. Life support devices or systems are intended to be implanted in the human body or to support and/or maintain and sustain and/or protect human life. If they fail, it is reasonable to assume that the health of the user or other persons may be endangered.